

# Automated Analyst Report

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Query: Analyse supplier performance across all product types. Identify which product categories have the highest defect rates, which shipping carriers offer the best revenue-to-cost ratio, flag any SKUs with failed inspection results, and highlight stockout risk where lead times are long relative to stock levels. report findings must be backed by solid proof from within the data

**\*\*EXECUTIVE SUMMARY\*\*** Haircare products exhibit the highest average defect rate (2.48 %) while Carrier C delivers the strongest revenue-to-shipping-cost ratio ( $\approx$  \$1,139 per shipping dollar). A total of 36 SKUs failed inspection, with three exceeding a 4.5 % defect rate, and several high-risk stockouts are evident—most notably SKU34 (risk score 13) and SKU68 (zero on-hand inventory). Immediate attention to these quality and inventory exposures is required to protect margins and service levels.

**\*\*KEY FINDINGS\*\*** - **\*\*Product type defect exposure:\*\*** Haircare = 2.483 % (highest), Skincare = 2.335 %, Cosmetics = 1.919 % (lowest). - **\*\*Carrier efficiency:\*\*** Carrier C generates \$1138.56 of revenue per shipping dollar (best), followed by Carrier B (\$1,055.70) and Carrier A (\$917). Carrier C also maintains a moderate avg. defect rate (2.60 %). - **\*\*Failed inspection SKUs:\*\*** 36 SKUs flagged; top three defect rates are SKU42 (4.94 %), SKU65 (4.91 %), SKU50 (4.75 %). SKU68 records a defect rate of 1.98 % but has **\*\*0 units in stock\*\***. - **\*\*Transportation mode quality:\*\*** Air mode shows the lowest avg. defect rate (1.824 %), outperforming Road (2.621 %), Rail (2.319 %) and Sea (2.315 %). - **\*\*Stock out risk:\*\*** SKU34 (skincare) scores 13 (stock = 1, lead time = 26 days); SKU68 (haircare) scores 8 with zero inventory; other high-risk SKUs have stock  $\leq$  5 and lead times  $\geq$  17 days. - **\*\*Margin resilience:\*\*** Gross margin ratios remain above 99 % for all product types (Cosmetics 0.993, Haircare 0.991, Skincare 0.992), indicating pricing strength despite quality and inventory pressures.

**\*\*RISKS AND RECOMMENDATIONS\*\*** The convergence of elevated defect rates in haircare, a concentration of failed inspection SKUs on Carrier C and Rail/Sea routes, and critical stockout scores (especially SKU34 and SKU68) threatens service reliability and margin protection. Prioritize a root cause review of the top defect SKUs, enforce stricter quality controls with Carrier C and Rail partners, and accelerate replenishment or safety stock adjustments for the highest risk items to mitigate potential lost sales and brand impact.